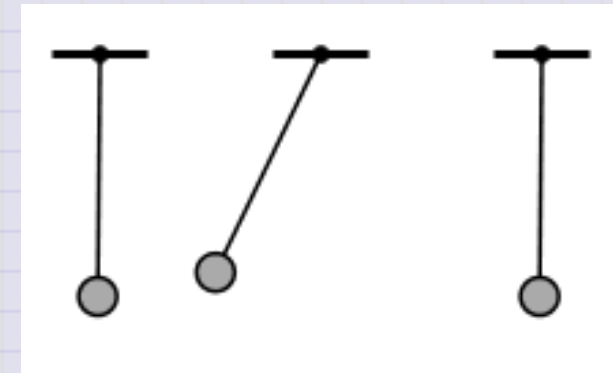


# Day 24 - Driven Oscillations

Resonance in a driven pendulum system.  $\longrightarrow$

Source: [Wikipedia](#)



# Announcements

- Midterm 1 is still being graded
- Homework 6 is due next Friday
- Homework 7 will be posted soon
  - Final project check-in #1
- Danny will be out this Wednesday
  - Class will be on zoom <https://msu.zoom.us/j/92295821308>
  - passcode: phy321
  - No office hours Wednesday

# Seminars this week

Most of MSU folks are at APS Global Physics Summit

## WEDNESDAY, March 19, 2025

- Astronomy Seminar, 1:30 pm, 1400 BPS, Alex Rodriguez, University of Michigan, *Galaxy clusters, cosmology, and velocity dispersion*
- FRIB Nuclear Science Seminar, 3:30pm., FRIB 1300 Auditorium, Dr. Pierre Morfouace of CEA-DAM, *Mapping the new asymmetric fission island with the R3B/SOFIA setup*

## THURSDAY, March 20, 2025

Colloquium, 3:30 pm, 1415 BPS, Guillaume Pignol, University of Grenoble, *Ultracold neutrons: a precision tool in fundamental physics*

# Reminders

We solved the damped harmonic oscillator equation:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$$

We chose a solution (**ansatz**) of the form

$$x(t) = C_1 e^{rt} + C_2 e^{rt}$$

and computed the roots of the characteristic equation:

$$r^2 + 2\beta r + \omega_0^2 = 0$$

We found the roots to be:

$$r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

# Weak Damping

We found that when  $\beta^2 < \omega_0^2$ , the roots are complex:

$$r = -\beta \pm i\sqrt{\omega_0^2 - \beta^2}$$

This means that the solution is oscillatory:

$$x(t) = e^{-\beta t} \left( C_1 \cos(\sqrt{\omega_0^2 - \beta^2} t) + C_2 \sin(\sqrt{\omega_0^2 - \beta^2} t) \right)$$

The solution is a damped oscillation with frequency  $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$ .

# Strong Damping

When  $\beta^2 > \omega_0^2$ , the roots are real:

$$r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

This means that the solution is not oscillatory:

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

where  $r_1 = -\beta + \sqrt{\beta^2 - \omega_0^2} < 0$  and  $r_2 = -\beta - \sqrt{\beta^2 - \omega_0^2} < 0$ .

The solution is the sum of two exponentials with different decay rates.

# Critical Damping

When  $\beta^2 = \omega_0^2$ , the roots are real and equal (repeated roots):

$$r = -\beta$$

This means that the solution is not oscillatory, but also that our ansatz is not sufficient.

The correct form of the solution is:

$$x(t) = (C_1 + C_2 t)e^{-\beta t}$$

**In most cases, we will work with weak damping.**

## Clicker Question 24-1

What do we expect the phase space diagram ( $x$  vs  $\dot{x}$ ) to look like for a weakly damped harmonic oscillator?

1. A set of ellipses
2. A set of spirals
3. Depends on how weak the damping is
4. Depends on the total energy
5. More than one of the above



## Clicker Question 24-2

The driven harmonic oscillator equation is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$$

with  $\omega_0^2 = k/m$  and  $2\beta = b/m$ . What is the dimension of the driving force  $f(t)$ ?

1. Force (Newtons, N)
2. Force per unit second (N/s)
3. Force per unit length (N/m)
4. Force per unit mass (N/kg)

## Clicker Question 24-3

The driven harmonic oscillator equation is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t).$$

This ODE is a \_\_\_\_\_ differential equation.

1. linear
2. nonlinear
3. first-order
4. second-order
5. more than one of the above

## Example: Sinusoidal Driving Force

Let  $f(t) = f_0 \cos(\omega t)$ , so that the driven harmonic oscillator equation is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t)$$

**Note:**  $\omega \neq \omega_0$

Note that if the driving follows a sine wave, then we have:

$$\ddot{y} + 2\beta\dot{y} + \omega_0^2 y = f_0 \sin(\omega t)$$

Interesting,  $e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$ , let try to work with  $z(t) = x(t) + iy(t)$ .

## Clicker Question 24-4

We found that the square amplitude of the driven harmonic oscillator is:

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

When is the amplitude of the driven oscillator maximized?

1. When the driving frequency ( $\omega$ ) is far from the natural frequency ( $\omega_0$ )
2. When the driving frequency ( $\omega$ ) is close to the natural frequency ( $\omega_0$ )
3. When the damping ( $2\beta$ ) is weak
4. When the damping ( $2\beta$ ) is strong
5. Some combination of the above